

Lab #8

NAMED PIPES.

Objective

Study the features of Named Pipes.

Exercises

1. What did you learn after running the above Program 1 and Program 2?

Running Program 1 and Program 2 demonstrates inter-process communication using a named pipe (FIFO) in Unix-like systems. Program 1 opens '/tmp/testfile' for reading and writing, continuously reading messages and printing them, while Program 2 writes messages passed as command-line arguments to the same pipe. Proper execution requires the named pipe to exist and both programs to run simultaneously, with Program 1 ideally starting first. Observations include the blocking nature of read and write operations, the need for error handling when the pipe is inaccessible, and the importance of message size management, ensuring messages fit within the defined buffer size (MSGSZ + 1). This exercise highlights the basics of named pipes, error handling in file operations, and synchronization between reading and writing processes for effective communication.

2. What did you learn after running Program 3?

Running Program 3 demonstrates the process of creating and using a named pipe (FIFO) for inter-process communication in Unix-like systems. The program starts by attempting to create the named pipe '/tmp/testfile' with read and write permissions using 'mknod'. If the creation fails, it prints an error message. It then opens the pipe for reading and writing, entering an infinite loop where it reads messages from the pipe and prints them to the console.